Introduction to Data Visualization

Why Visualization Skills Matter in Careers?

01

Critical for roles like Data Analyst, BI Analyst, and Data Scientist.

02

Enhances your ability to present data-driven insights.

03

Visualization tools (e.g., Tableau, Power BI) are tested in interviews.

Learning Objectives

By the end of this session, you will be able to:

- Understand **the importance of data visualization.**
- Understand the various **visualization techniques.**
- Learn how to **create and interpret** the various plots in Python.

Data Visualization in Decision-Making



Helps stakeholders understand trends and patterns quickly.



Increases the impact of presentations and reports.



Critical for strategy formulation and operational insights.

How does Data Visualization help?

Transforms raw data into clear, actionable insights.

Enhances decision-making by making patterns easier to detect.

Key for storytelling and communicating findings effectively.

Common Visualization Techniques



Line Charts: Best for tracking data over time (e.g., COVID cases).



Bar Charts: Effective for comparing categories.



Pie Charts: Visualizing part-to-whole relationships.



Scatter Plots: Showing relationships between variables.

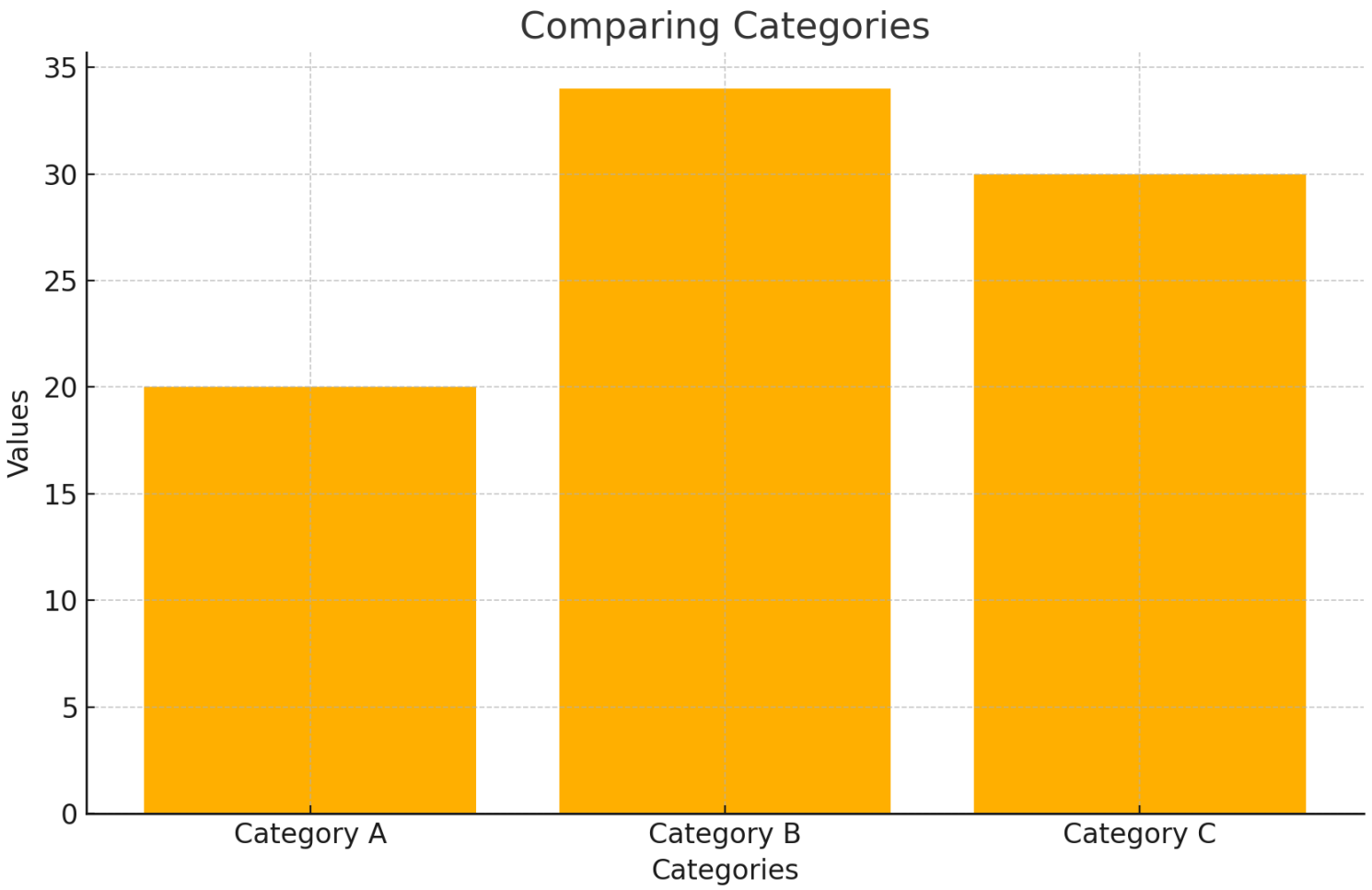


Heat Maps: Visualizing density or intensity of data.

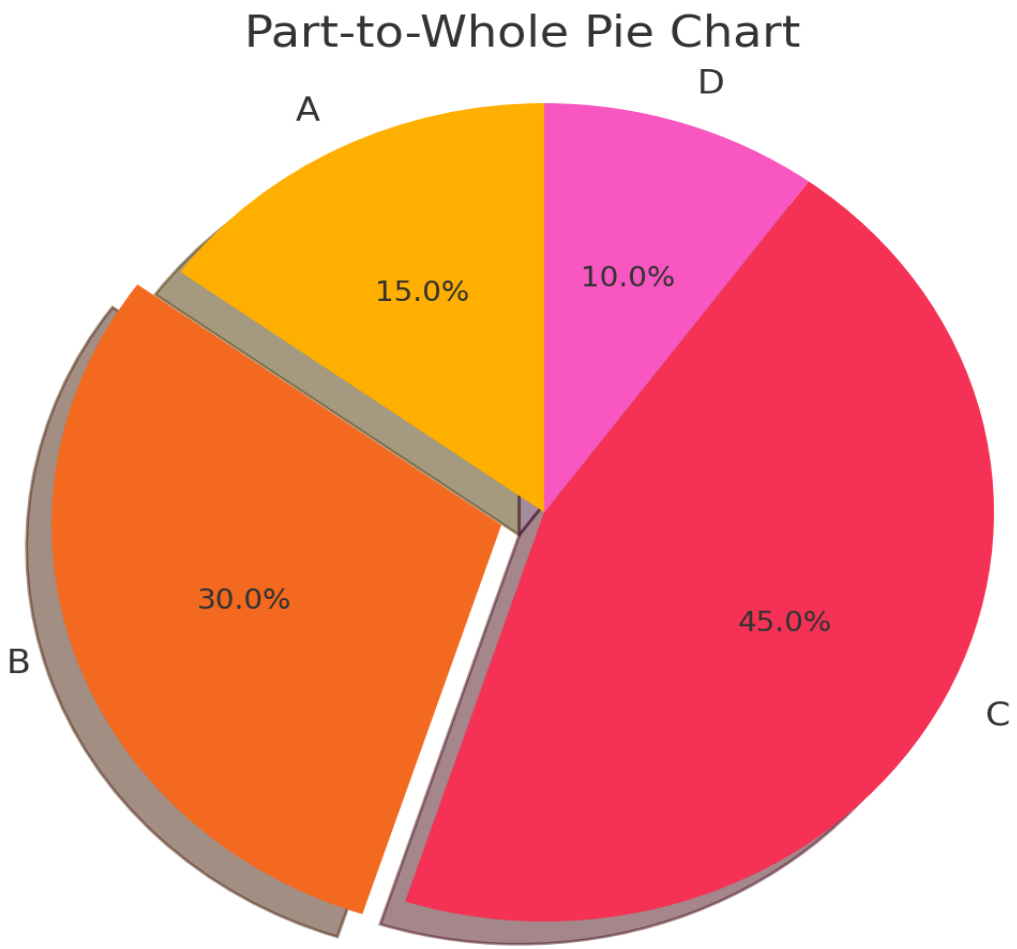
Line Chart Example: Stock Price of Bata India as on Sep 9



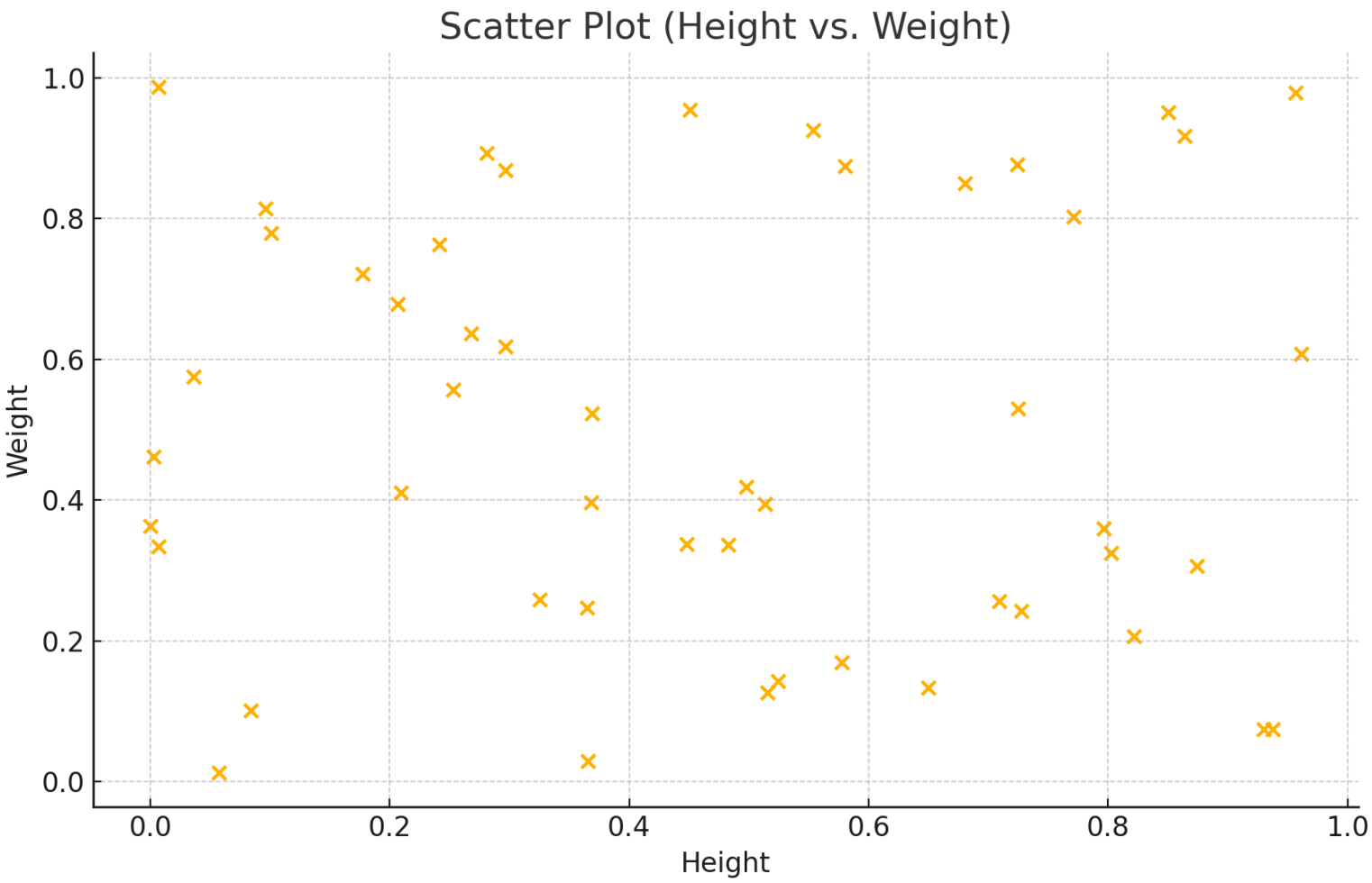
Bar Chart Example: Comparing Categories



Pie Chart Example: Part-to-Whole



Scatter Plot Example: Height vs Weight



How to Choose the Right Visualization

1. Identify the type of data you are working with.
2. Understand the audience and the message you want to convey.
3. Choose charts that simplify complex data.
4. Avoid over-complicating with too many visual elements.

Multiple Choice Questions (MCQs)

1. A data scientist wants to show the monthly trend of website traffic over a year. Which visualization is most appropriate?

A. Scatter Plot
B. Line Chart
C. Pie Chart
D. Bar Plot

2. A researcher wants to examine the relationship between students' study hours and their exam scores. Which visualization should be used?

A. Pie Chart
B. Line Chart
C. Scatter Plot
D. Bar Plot

Sample Questions

Sample Questions:

- "How would you approach visualizing sales performance across regions?“
- "Can you explain a time when a visualization led to actionable business insights?"
- Hint: Practice explaining your thought process in both technical and business terms.

Python Implementation

- ‘mpg’ dataset in seaborn
- You are a data scientist working at car sales outlet and you have been asked to perform EDA using visualization on the data….

	mpg	cylinders	displacement	horsepower	weight	acceleration	model_year	origin	name
0	18.0	8	307.0	130.0	3504	12.0	70	usa	chevrolet chevelle malibu
1	15.0	8	350.0	165.0	3693	11.5	70	usa	buick skylark 320
2	18.0	8	318.0	150.0	3436	11.0	70	usa	plymouth satellite
3	16.0	8	304.0	150.0	3433	12.0	70	usa	amc rebel sst
4	17.0	8	302.0	140.0	3449	10.5	70	usa	ford torino

Session Summary

In this session, we have discussed about:

- The **importance of visualization** from career as well as business perspectives.
- The various **visualization techniques** and when to us each one of these.
- How to **create and interpret** the various plots in Python.

Reference Links

- How to Create Stunning Data Visualizations in Python: Top 10 Techniques to Learn
- Mastering Credit Risk Analysis: A Step-by-Step Guide to Descriptive Statistics in Python
- How Much Statistics Do You Really Need for Data Science?
- Phases of data science and analytics.